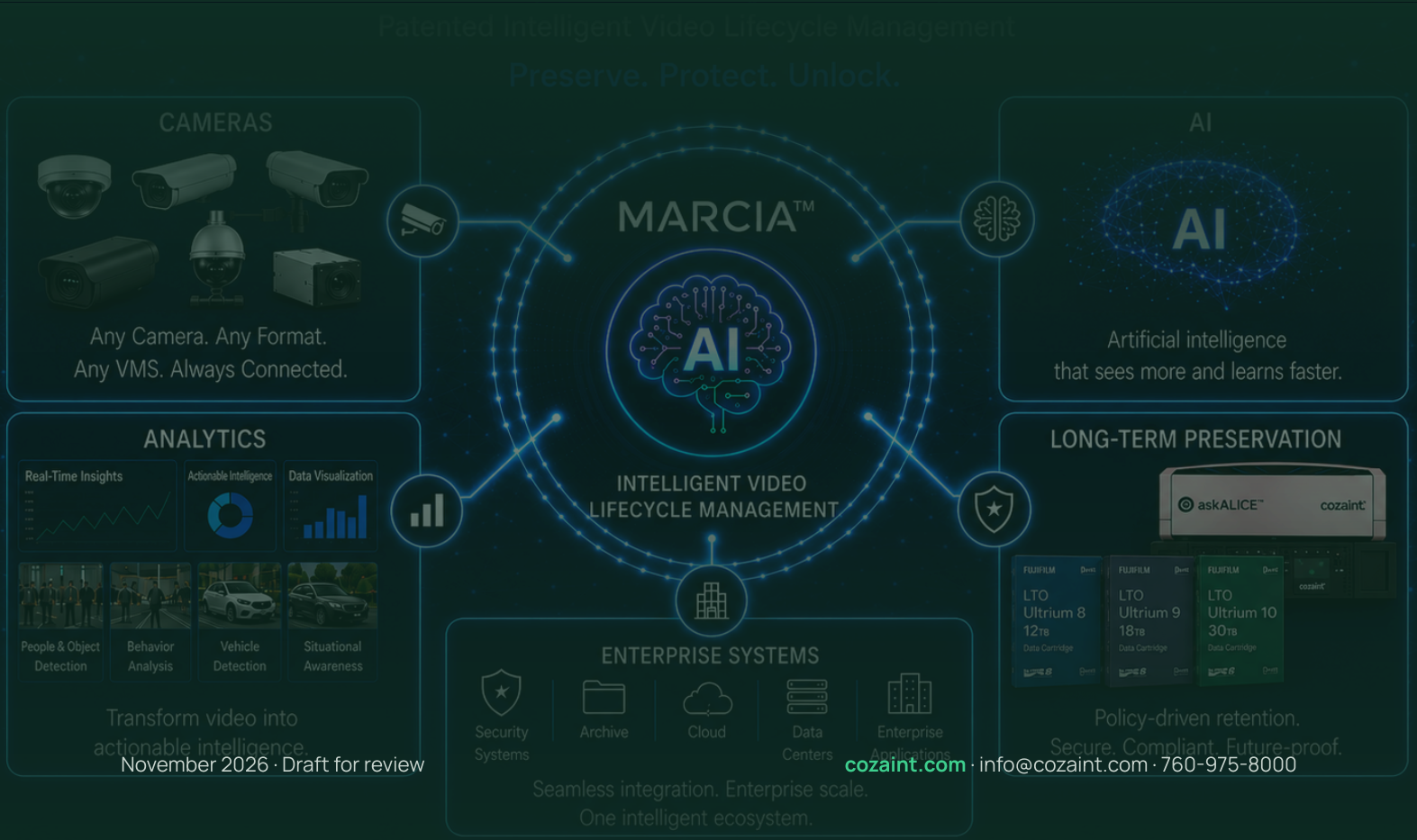
cozaint®

TECHNICAL OVERVIEW

MARCIA™ Architecture & Patent Overview

What Cozaint invented, how it works, and why it makes LTO practical for the surveillance operator



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MARCIA in one paragraph

MARCIA™ Intelligent Video Lifecycle Management is the layer between a video management system (VMS) and long-term storage. It keeps recent video on disk where the VMS already records it, moves retained video to LTO tape under policy, and, when an operator asks for older video on the normal VMS timeline, brings it back from tape to that same timeline. The operator never leaves the VMS, never files a restore request and never needs IT. That timeline-driven recall is protected by **U.S. Patent No. 12,755,902**.

The patent at a glance. U.S. Patent No. 12,755,902, issued October 6, 2026 to Cozaint Corporation (application No. 18/065,506). Its patented architecture lets an operator's normal VMS playback request recall archived video from LTO and play it on the same timeline they use every day.

Why an invention was needed

LTO has been the archive of choice in Media & Entertainment, research and enterprise IT for decades. It never caught on in surveillance because of the workflow, not the media:

- **Operators do not know in advance what they need.** Pre-fetching video from tape assumes they do.
- **Backup-and-restore belongs to IT.** It forces the operator out of the VMS and into a ticket queue.
- **The VMS timeline has no end.** Unlike an editor's finite project, surveillance history grows every second, so the archive must answer requests for any camera, any date and any time.

Competitors can sell tape libraries, hierarchical storage software or disk archives. None of those, on their own, make the archive part of the operator's normal workflow. That is what MARCIA does, and that is what the patent protects.

The patented workflow

Request.
The operator moves the VMS timeline to a past date and time.

Locate.
MARCIA identifies which storage tier holds that video.

Map.
It maps the request to the exact cartridge and position.

Recall.
The tape library loads the cartridge and reads the video.

Play.
The video returns to the same VMS timeline.

What happens at each step

Step	What the operator sees	What MARCIA does
1 Request	Moves the timeline, exactly as for recent video	Receives the VMS playback request for that camera and time
2 Locate	Nothing new	Determines whether that video is on the disk tier or the LTO tier
3 Map	Nothing new	Maps the camera and time to the specific cartridge and position where it was written
4 Recall	A visible retrieval status	Directs the library to load the cartridge into a drive and reads the requested video
5 Play	The video plays on the same timeline	Returns the original recording to the VMS for playback and export

If the video is still on disk, steps 3 and 4 are skipped and playback starts at once. If it is on LTO, recall typically takes a few minutes, depending on the library, the drives and where the video sits on the cartridge. No extra steps, no separate application, no IT ticket.

Because the operator works with the original recording rather than a re-encoded copy, the workflow supports chain of custody when video is exported as evidence.

The architecture

Three stages of the video lifecycle

Stage	What happens	Where the video lives
01 Active video	The VMS records and plays recent video as it always has	Disk tier on the recording server
02 Governed retention	Retention rules decide what moves to LTO and when; new video is replicated to tape automatically in the background	LTO cartridges in the library (and offline if exported)
03 Connected workflow	Requested history returns to the operator through the patented workflow	Recalled from LTO to the VMS timeline

Building blocks

- **VMS.** The customer's existing video management system remains the operator's only interface. MARCIA works with Milestone XProtect, Hanwha WAVE, Digital Watchdog DW Spectrum, Network Optix Nx Witness and Cozaint BOBBYvms.
- **Recording server with a disk tier.** Sized for the recent video operators review most often; the archive does not have to fit on disk.
- **MARCIA.** Applies the retention policy, tracks where every camera's video is stored on tape, answers recall requests and reports retrieval status.
- **LTO tape library.** From a compact 2U library to a full 42U rack and beyond. MARCIA is validated on HPE StoreEver and Quantum Scalar libraries and ATTO XstreamCORE.
- **Connection.** In a typical deployment the server connects to the library through a SAS host bus adapter (HBA) and an external SAS cable; Fibre Channel is an option for larger installations. The *How It All Connects* guide shows the physical setup step by step.

Design choices that matter in practice

Choice	Why it matters
Video for each camera kept together on tape	Recalling one camera's history touches as few cartridges as possible
At least two LTO drives (best practice)	One drive can serve a recall while another keeps writing new video
Policy-driven movement	Retention follows the organization's rules, not an operator's memory
Visible request and retrieval status	Operators know their video is on the way, so they do not re-request or call IT
Open LTO format (LTFS)	Cartridges are self-describing and readable on standard systems; the archive is not locked to one server
Offline cartridges	Cartridges can be exported and stored offline, giving an air-gapped copy outside the reach of the network
WORM media where required	Write-once cartridges can be used where policy calls for recordings that cannot be overwritten

Being precise about protection. An LTO cartridge is air-gapped only when it is removed from the library or otherwise disconnected; cartridges in the library are online. Recordings are protected from being overwritten when written to WORM media or governed by retention policy. Design each deployment to the level of protection the customer's policy requires.

Scaling

Retention grows by adding cartridges and, as needed, library modules and drives, not by adding racks of spinning disk. LTO-10 stores 30 TB per cartridge (up to 40 TB with the newest LTO-10 cartridges), and the LTO roadmap extends to LTO-14 at a planned 576 TB per cartridge. The same architecture runs from a single compact library to enterprise libraries holding thousands of cartridges.

One platform, three ways to deploy

	MARCIA™	askALICE™	ALICE1™
What it is	The Intelligent Video Lifecycle Management platform	Enterprise turnkey disk-plus-LTO system	Compact LTO archive appliance
Powered by	–	MARCIA	MARCIA
Best for	Enterprises adding a retention tier to the VMS they run	Large sites that want one complete system	Smaller and self-contained sites
Scale	Hundreds of terabytes to petabytes	Hundreds of terabytes to petabytes	Single appliance with integrated disk cache and LTO drive

Where MARCIA fits best

- Long or mandated retention: 6 to 12 months and beyond
- Hundreds or thousands of cameras, high resolution, high frame rates
- Storage needs of roughly 400 TB and up, or a disk refresh coming due
- Litigation exposure, regulatory retention or evidence policies
- Concern about power, cooling, rack space or ransomware, or an existing Milestone, Hanwha, Digital Watchdog or Network Optix environment

Media & Entertainment proved LTO for video. MARCIA makes it practical for surveillance. Patented VMS integration makes the storage tier invisible to the operator. The result is substantially more retained video at dramatically lower lifecycle cost.

Talk to an engineer. Request a MARCIA assessment at **cozaint.com/marcia**, or contact **info@cozaint.com** · **760-975-8000**. Cozaint Corporation, headquarters Palmetto Bay, Florida, office San Diego, California.